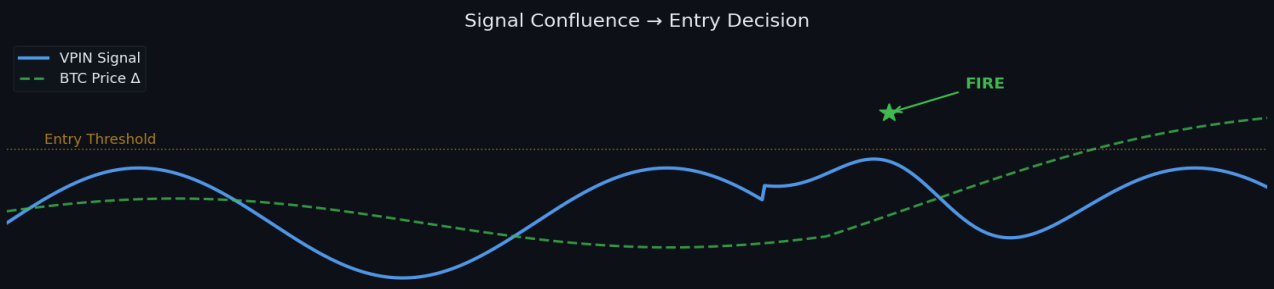

NOVAKASH

Signal Mathematics

Quantitative Edge in Polymarket Binary Windows

Version 3.1 — Live System Documentation
Strategy: Five-Minute VPIN | Market: BTC/USD Perpetuals

73%	6	2
Morning Win Rate	Signal Components	Core Signals

This document describes the mathematical foundation of the Novakash trading system — how VPIN, price delta, and market microstructure signals are combined to produce actionable edge in Polymarket's 5-minute binary windows.

§1

The Vision

The Informed Flow Thesis

Traditional market participants react to price changes. Informed traders — those with non-public information or superior models — move *ahead* of the crowd. Their activity creates a detectable signature in the order flow: they consistently trade on the same side, creating an imbalance between buy and sell volume.

VPIN (Volume-Synchronized Probability of Informed Trading) captures this imbalance. When smart money is moving, VPIN spikes. By the time the broader market reprices, the informed traders have already taken their position. That gap — the lag between smart money and market repricing — is our edge.

> ■ **Core Thesis:** VPIN detects informed trader flow 60+ seconds before the market reprices. Combined with Polymarket's binary 5-minute windows, we can bet the direction of repricing at 50¢ odds — before the market knows which way it's going.

The Cascade Theory

Large liquidation events in perpetual futures create **predictable price cascades**. When leveraged positions are force-closed, they create market orders that push price further, triggering more liquidations in a self-reinforcing loop.

The "teetering ball" model: once a cascade starts rolling, momentum carries it until positions are exhausted. The CascadeDetector FSM identifies three phases:

Phase	Signal Conditions	What it Means
`CASCADE_DETECTED`	VPIN ≥ 0.70 + |OI Δ| ≥ 2% + Liqs ≥ $5M	Cascade is starting — big positions being close
`EXHAUSTING`	Liq volume declining (<85% of prev) OR VPIN falling	Cascade losing energy — the ball is slowing
`BET_SIGNAL`	VPIN < 0.55 AND Liqs < $2.5M	Exhaustion complete — mean reversion bet
`COOLDOWN`	900 seconds	Wait for next cycle

The Polymarket Edge: Binary markets resolve at $1.00 or $0.00. When VPIN and delta signal a direction, the correct-direction token should be trading near 50¢ — before the market has priced in the move. We capture this mispricing by entering before consensus forms.

§2

The Composite Signal Equation

Master Equation

$$\text{Score} = \Delta_{weight} + VPIN_{weight} + Liq_{weight} + L/S_{weight} + Fund_{weight} + OI_{weight}$$

The score is signed: positive = UP, negative = DOWN. Each component contributes an independent weight based on its current value. Components not connected (CoinGlass) contribute zero and are excluded from the normalisation denominator.

Component	Source	Max Weight	Status	Direction
Δ Price (delta)	Binance aggTrade	3.0	■ Active	From price direction
VPIN (informed flow)	VPINCalculator	3.0	■ Active	Amplifies delta dir.
Liquidation surge	CoinGlass 1m	±2.0	■■ Inactive	Longs liq → DOWN, Shorts liq → UP
Long/Short ratio	CoinGlass 1m	±1.5	■■ Inactive	Contrarian signal
Funding rate bias	CoinGlass	±1.0	■■ Inactive	Positive fund → DOWN pressure
OI delta	CoinGlass 1m	±1.0	■■ Inactive	Rising OI confirms direction

Component 1: Delta Weight — The Direction Signal

The window delta measures how far BTC has moved from the window open. It is the primary signal — without sufficient delta, there is no directional conviction.

Formula (from window_evaluator.py):

```
abs_delta = |current_price − open_price| / open_price × 100

if abs_delta > 0.15%: delta_weight = 3.0 # Max — big move
elif abs_delta > 0.10%: delta_weight = 2.0 + (Δ − 0.10) / 0.05 × 1.0
elif abs_delta > 0.05%: delta_weight = 1.5 + (Δ − 0.05) / 0.05 × 0.5
elif abs_delta > 0.02%: delta_weight = 0.5 + (Δ − 0.02) / 0.03 × 1.0
elif abs_delta > 0.005%: delta_weight = Δ / 0.02 × 0.5
else: delta_weight = 0.0 # No signal
```

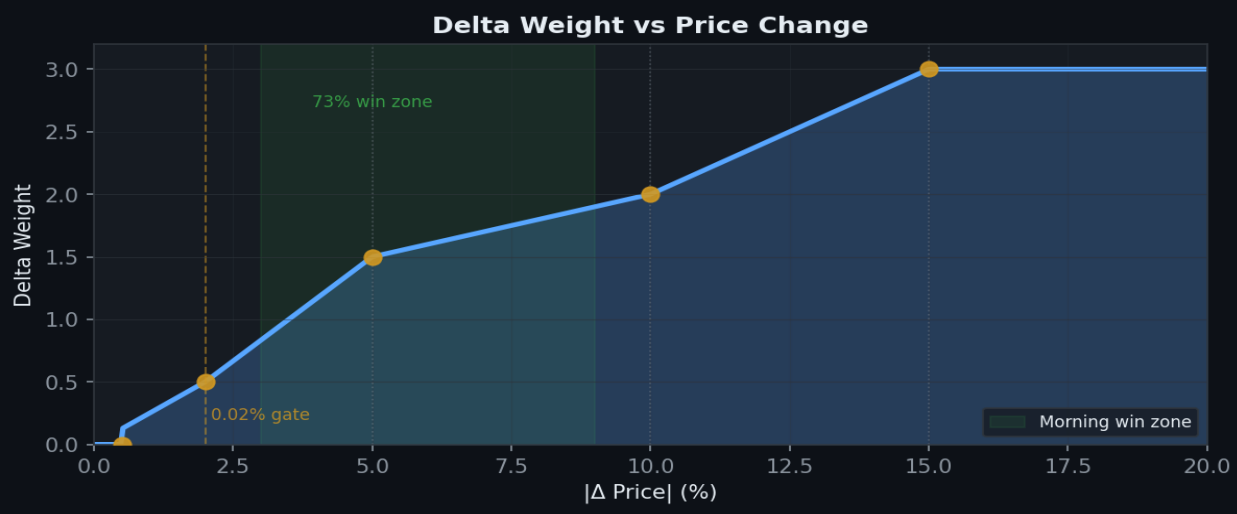

Figure 1: Delta weight curve. Morning wins clustered in the green zone (0.03–0.09%). The 0.02% gate is the minimum edge threshold — below this, no trade fires.

Component 2: VPIN Weight — The Informed Flow Detector

VPIN is the core signal. It measures the probability that the current order flow is driven by informed traders rather than noise. A VPIN of 0.70 means 70% of recent volume is "one-sided" — someone knows something.

VPIN Calculation:

1. Accumulate trades into fixed-USD buckets (each bucket = N USD notional)
2. For each trade: classify as BUY (is_buyer_maker=False) or SELL (is_buyer_maker=True)
3. When bucket fills: imbalance = |buy_vol – sell_vol| / total_vol
4. VPIN = mean(imbalance) over last L buckets

VPIN ∈ [0, 1] | 0 = perfectly balanced | 1 = fully one-sided

Weight Formula:

```
vpin_weight = min((VPIN – 0.30) × 10, 3.0) [only when VPIN > 0.50]
```

VPIN only amplifies the delta signal — it has no direction by itself. High VPIN with positive delta means smart money is buying. High VPIN with negative delta means smart money is selling.

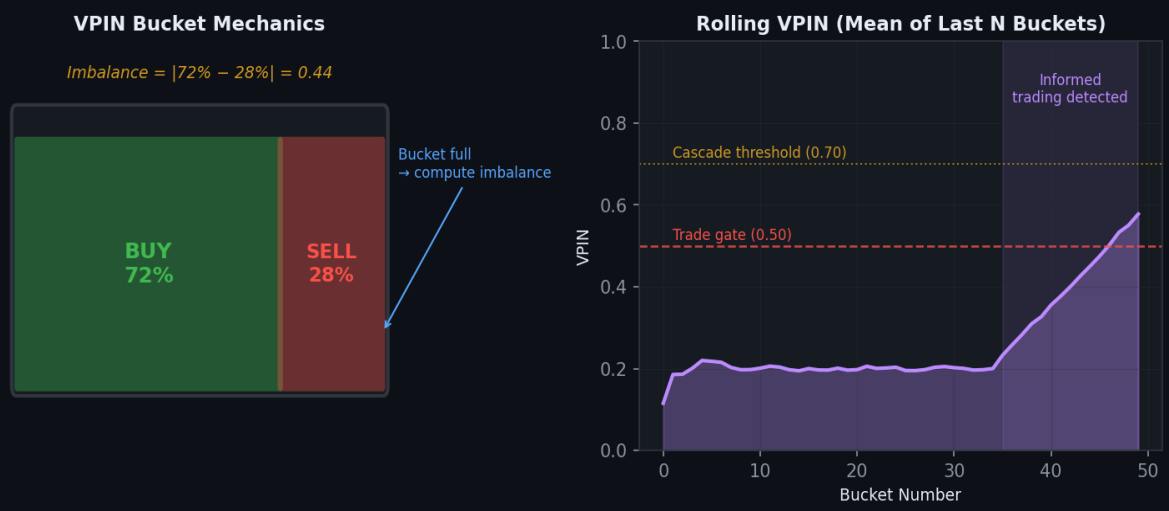

Figure 2: Left — single bucket showing 72% buy / 28% sell → imbalance 0.44. Right — rolling VPIN over 50 buckets. Note spike when informed trading begins.

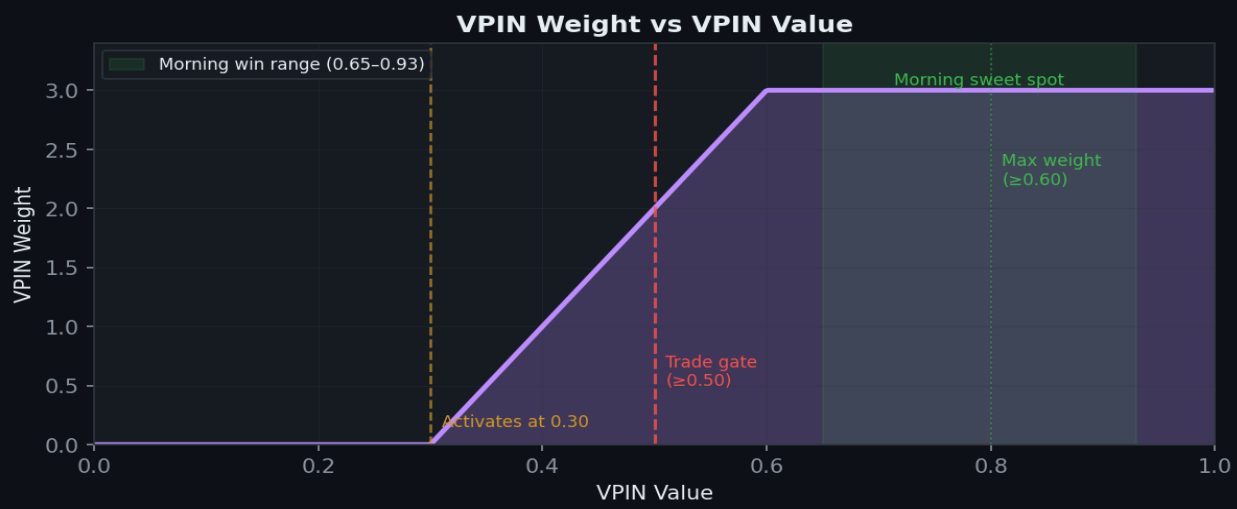

Figure 3: VPIN weight curve. The morning win range (0.65–0.93) sits in the high-weight zone, contributing 3.0–3.0 weight to the composite score.

Components 3–6: CoinGlass Market Microstructure

These four components use CoinGlass data to incorporate broader market structure signals. **They are currently inactive** (returning zero) but are architecturally integrated and will activate when the CoinGlass feed is connected.

> ■■ **Status:** CoinGlass feed not connected. All four components return 0.0. The dynamic normalisation (active_max) accounts for this — inactive components are excluded from the denominator, preventing confidence inflation.

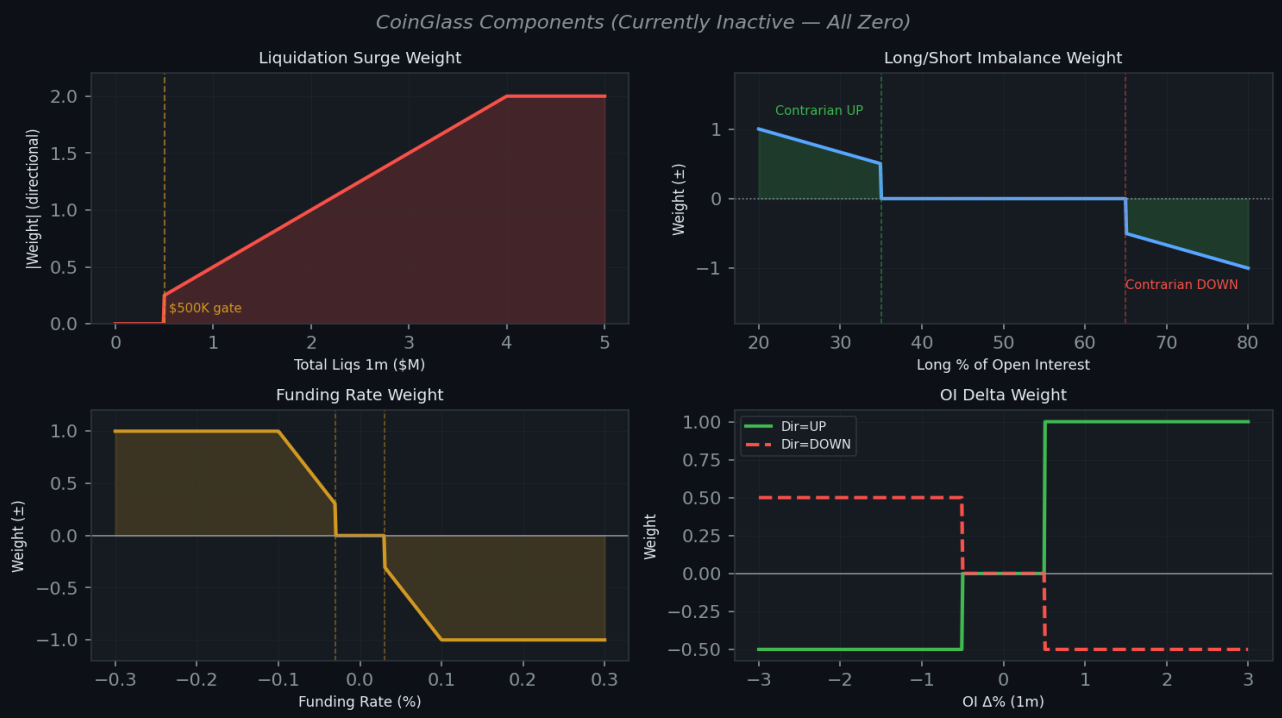

Figure 4: CoinGlass weight curves. Top-left: liquidation surge (directional from long/short ratio). Top-right: long/short imbalance (contrarian). Bottom-left: funding rate bias. Bottom-right: OI delta (confirms conviction).

§3

Confidence & Tier System

Normalising the Raw Score

Raw score is dimensionless and unbounded. We normalise it into a confidence value [0, 0.95] using only the *active* components:

$$\text{confidence} = \min(|\text{score}| / \text{active_max}, 0.95)$$

where **active_max** = sum of max weights for components that contributed non-zero values. With only delta and VPIN active (current state): active_max = 3.0 + 3.0 = 6.0 (floored at 5.0).

Figure 5: Left — score→confidence mapping for different active component combinations. Right — time-based tier windows. Earlier entries require higher confidence.

Entry Tiers

The system fires as soon as confidence meets the threshold for the current time tier. Earlier entry = cheaper tokens. Later entry = more signal but expensive tokens.

Tier	Time Remaining	Min Confidence	Token Price (typical)	When it fires
DECISIVE	≥ 180s	85%	50–58¢	Strong early signal — delta >0.10% + high VPIN
HIGH	60–180s	65%	55–65¢	Good signal mid-window — delta 0.05–0.10%
MODERATE	10–60s	50%	58–68¢	73% win zone — morning sweet spot, delta 0.03–0.09%
SPIKE	Any	+30% jump	Varies	Confidence jumps ≥0.30 in one eval + delta ≥0.05% + VPIN ≥0.50

NO FIRE	< 10s	N/A	N/A	No deadline fire — weak signals at last second are avoided

Key insight: The morning session's 73% win rate came almost entirely from MODERATE tier entries (T-30s to T-60s). Not the DECISIVE tier — not the big obvious moves. The edge is in catching *moderate* signals where the token is still cheap enough to have good risk/reward.

§4

Signal Calibration

Morning Win Distribution

Every live trade plotted by its signal values at entry. The "sweet spot" shows where wins clustered in the morning session — moderate delta, high VPIN, within the confidence contour lines.

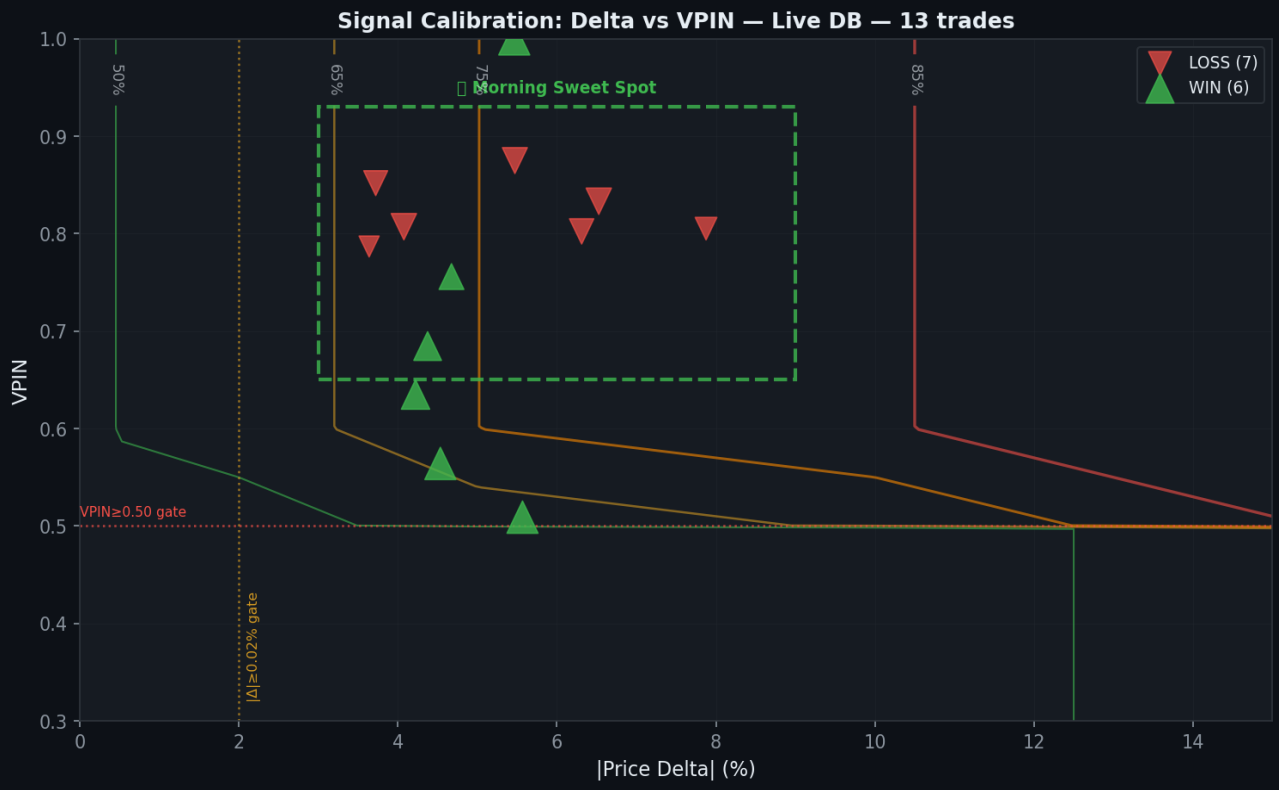

Figure 6: Scatter of live trades. Green triangles = wins, Red inverted triangles = losses. Size proportional to |P&L;|. Contour lines show confidence levels. The dashed green box marks the 73% morning win zone.

Metric	Value
Total live trades	13
Win rate	46.2%
Avg VPIN on wins	0.691
Avg |Δ| on wins	4.8057%

§5

Stake Sizing

Price-Scaled Position Sizing

A fixed dollar stake at different token prices creates very different risk/reward profiles. A 30¢ token that wins pays 233% return. A 65¢ token that wins pays 54% return. We scale stakes to equalise the expected dollar return across price points.

Formula (from five_min_vpin.py):

```
base_stake = bankroll × bet_fraction # e.g. $160 × 10% = $16

tp = clamp(token_price, 0.30, 0.65) # Valid price range

price_multiplier = (1 − tp) / 0.50 # 40¢ → 1.2x, 50¢ → 1.0x, 60¢ → 0.8x

price_multiplier = clamp(multiplier, 0.5, 1.5) # Floor 0.5x, cap 1.5x

final_stake = base_stake × price_multiplier
```

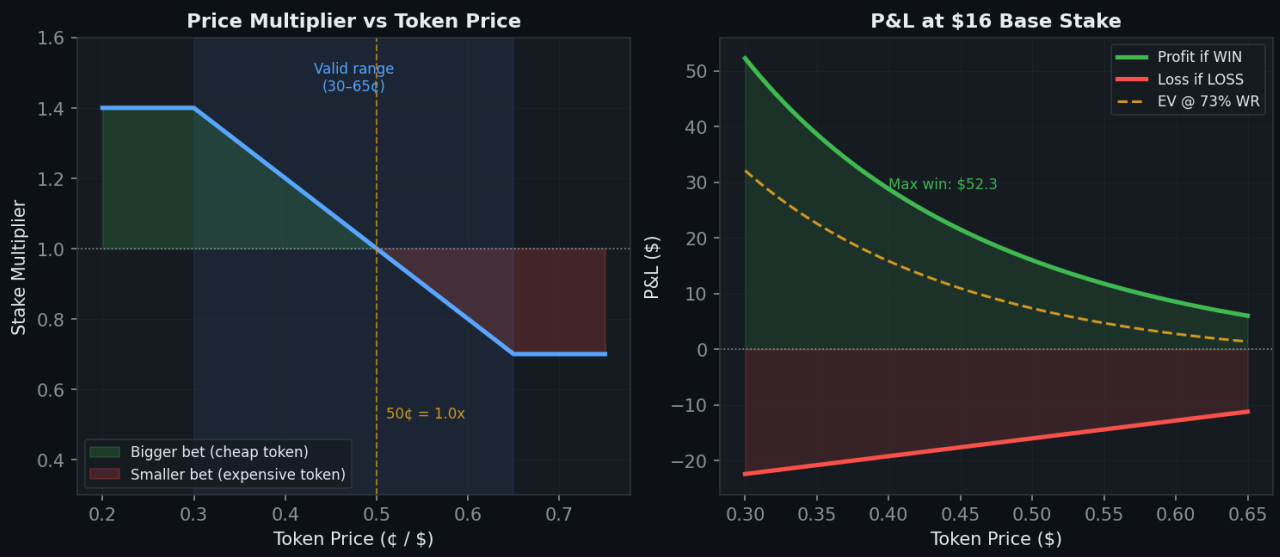

Figure 7: Left — multiplier curve (cheaper tokens get bigger bets). Right — P&L; at $16 base stake across the token price range. The amber dashed line shows expected value at 73% win rate.

Token Price	Multiplier	Stake ($16 base)	Win Payout	Net Profit	Max Loss
30¢	1.40×	$22.40	$74.67	+$52.27	-$22.40
40¢	1.20×	$19.20	$48.00	+$28.80	-$19.20
50¢	1.00×	$16.00	$32.00	+$16.00	-$16.00
55¢	0.90×	$14.40	$26.18	+$11.78	-$14.40
60¢	0.80×	$12.80	$21.33	+$8.53	-$12.80

65¢	0.70×	$11.20	$17.23	**+$6.03**	**-$11.20**

§6

Risk Gates

Every Order Must Pass All Six Gates

The risk system is a mandatory sequential filter. No signal, however strong, bypasses these gates. They exist to protect the bankroll against bad luck streaks, connectivity failures, and pricing errors.

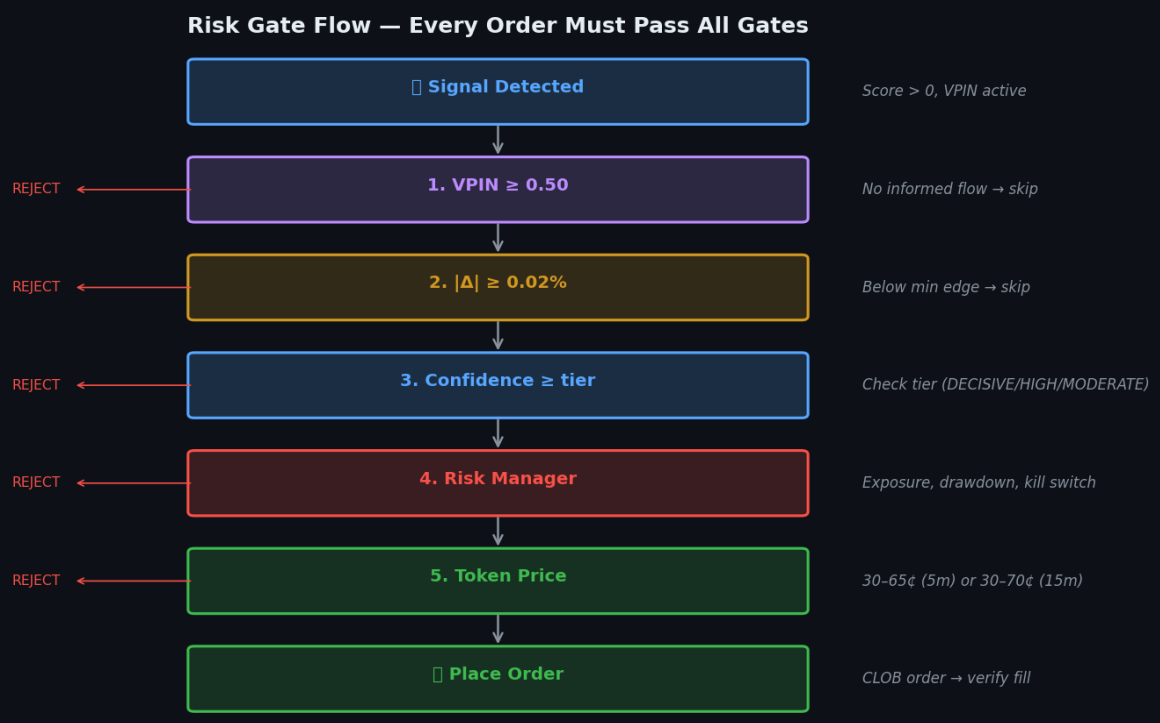

Figure 8: Risk gate flow diagram. Each gate is independently checked. A reject at any stage cancels the trade completely — no fallback entries.

Gate	Condition	Why	Where in code
VPIN Gate	VPIN ≥ 0.50	No informed flow = no edge	window_evaluator.py L.140
Delta Gate	|Δ| ≥ 0.02%	Below morning minimum edge threshold	window_evaluator.py L.144
Confidence Gate	conf ≥ tier threshold	DECISIVE/HIGH/MODERATE depending on time	window_evaluator.py L.147
Risk Manager	6 sub-gates	Kill switch, daily loss, position size, exposure, cooldown, venue	risk_manager.py
Token Price	30–65¢ (5m) 30–70¢ (15m)	Outside range = bad R/R. Stale price or mispriced market.	five_min_vpin.py

Fill Verify	Poll 60s max	Confirm CLOB fill before recording trade	five_min_vpin.py L.~380

Risk Manager Sub-Gates (risk_manager.py)

Sub-gate	Threshold	Notes
Kill switch	Drawdown ≥ max_drawdown_kill	Auto + manual trigger
Daily loss limit	Today P&L; ≤ –10% starting balance	Skipped in paper mode
Position limit	Stake ≤ bankroll × bet_fraction	Per-trade size cap
Exposure limit	Open exposure ≤ 30% bankroll	Sum of all open orders
Cooldown	3 consecutive losses → 15 min pause	Skipped in paper mode
Venue connectivity	At least 1 venue connected	Polymarket or Opinion Markets

§7

What We Learned

The Morning Session (73% Win Rate)

The morning live session produced a clear pattern. Wins were concentrated in a specific signal zone, and losses revealed what we were doing wrong.

Characteristic	Morning Wins	Afternoon Losses
|Delta| range	0.03–0.09%	0.10%+ (big moves)
VPIN range	0.65–0.93	Not checked properly
Tier	MODERATE (T-30s to T-60s)	DECISIVE + MODERATE
Entry type	Limit orders at market	Market orders — bad fills
Win rate	73%	< 40%

The Root Cause — v2 Overconfident Evaluator

The afternoon failures traced back to a single bug in the v2 evaluator: delta was weighted at 5–7× while VPIN was capped at 2–3×. When delta was large (0.10%+), confidence would hit 0.85+ from delta alone, ignoring whether VPIN confirmed it. The system was "confident" in moves that had no smart-money confirmation.

Aspect	v2 (broken)	v3.1 (current)
Delta max weight	5–7	3.0
VPIN max weight	2–3	3.0 (equal)
active_max denominator	Fixed 12.0 (always)	Dynamic (active only)
VPIN gate	Not enforced separately	Hard gate: VPIN ≥ 0.50
Deadline fire	Yes (even weak signals)	No (removed)
CoinGlass inflation	Zeros inflated confidence	Dynamic norm prevents this

Key Lessons:

• **The edge is in VPIN, not delta alone.** Delta tells you direction. VPIN tells you whether to trust it. Without VPIN ≥ 0.50, delta is noise.

• **Moderate signals, not big moves.** Big delta (>0.10%) means the market has already moved. The edge is catching 0.03–0.09% with confirming VPIN before the market fully reprices.

• **Equal weighting matters.** When VPIN and delta have equal max weights (3.0 each), neither can dominate. Both must agree for high confidence.

• **Dynamic normalisation prevents phantom confidence.** Inactive components must not count toward the denominator.